Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 28

The Principle of Mathematical Induction

Covers Chartrand & Zhang, Section 4.1

Status of this set

Ungraded practice. Solutions released with it. Every proof must have a labelled base step and a labelled inductive step.

Part I: How to Practice This Material

The template

1. "We proceed by induction."
2. **Base step:** check $P(1)$ by direct computation.
3. **Inductive step:** "Assume $P(k)$ for a positive integer k. We show $P(k+1)$." Write out $P(k+1)$ explicitly so you know the target.
4. **Use the hypothesis** — and say that you are using it.
5. "By the Principle of Mathematical Induction, $P(n)$ holds for every positive integer n."

For sums: split off the last term

(sum to $k+1$) = (sum to k) + (term $k+1$). The first piece is what the hypothesis describes. Substitute, then simplify toward the target.

Three mistakes to expect

1. **Assuming** $P(k+1)$ and working backward. You may assume only $P(k)$.
2. **Skipping the base step.** Lecture 28 showed a false claim with a valid inductive step.
3. **Never using the hypothesis.** If $P(k)$ is not substituted anywhere, it is not an induction proof.

Part II: Practice Problems

Section A — Sums

1. Prove that $2 + 6 + 10 + \cdots + (4n - 2) = 2n^2$ for every positive integer n.

2. Prove that $3 + 7 + 11 + \cdots + (4n - 1) = n(2n + 1)$ for every positive integer n.

3. Prove that $1 + 5 + 5^2 + \cdots + 5^n = \dfrac{5^{n+1} - 1}{4}$ for every positive integer n.

Section B — The base step matters

4. Consider the claim: $n^2 + n + 3$ *is even for every positive integer* n.

(a) Show that the inductive step can be carried out.

(b) Evaluate $n^2 + n + 3$ at $n = 1$.

(c) What do (a) and (b) together tell you?

Section C — Writing

5. In five to eight complete sentences, explain why induction proves a statement about *every* positive integer even though it checks only one case directly. Connect your answer to the Week 1 observation that checking cases is not proof.

Section D — Challenge (optional)

6. Compute $5 + 9 + 13 + \cdots + (4n + 1)$ for $n = 1, 2, 3, 4$, conjecture a closed formula, and prove it by induction.